

ECON 0100 | Fall 2024

MiniExam D | Version 2

This MiniExam will take 15 minutes with quick break to follow. MiniExams are designed to both test your knowledge and challenge you to apply familiar concepts in new environments. Treat it as if you're trying to show me that you understand the material. Answer clearly and completely.

Academic Conduct Code | Name: TAYLOR WEIDMAN Student ID:

The following academic conduct code is designed to protect the integrity of your work. Print your name/initials beside the three academic honesty agreements. I pledge to my fellow students, the university, and the instructor, that:

- ___ I will complete this MiniExam solely using my own work.
___ I will not use any digital resources unless explicitly allowed by the instructor.
___ I will not communicate directly or indirectly with others during the MiniExam.

ACC :: 10 pts

Question 1 (of 2) | Hermione's Potion Ingredients

Hermione and Draco share a garden plot growing moonflowers for their potions project. Each can harvest 1 or 2 flowers weekly. More harvesting reduces future growth. The payoff matrix shows their utility from different harvesting choices.

		Draco	
		H1	H2
Hermione	Harvest 1 (H1)	10, 10	6, 15
	Harvest 2 (H2)	14, 6	8, 8

(1)

Q1.a | Best Response 12 pts

What is Hermione's best response when Draco chooses to harvest 1? H2

What is Hermione's best response when Draco chooses to harvest 2? H2

What is Draco's best response when Hermione chooses to harvest 1? H2

What is Draco's best response when Hermione chooses to harvest 2? H2

FOR EACH:

correct: 3 pts

incorrect: 1 pts

Q1.b | Nash Equilibrium 12 pts

What is/are the Nash Equilibrium/Equilibria? Circle all that apply: (H1,H1), (H1,H2), (H2,H1), (H2,H2)

Q1.c | Social Efficiency 14 pts

What is the socially efficient outcome? Circle one: (H1,H1), (H1,H2), (H2,H1), (H2,H2)

14

100% :: correct: (H1, H2)

11% :: incorrect: (H2, H1)

2 :: incorrect

Question 2 (of 2) | Moonflower Management

In this question we will consider whether to impose a harvesting restriction. Five prefects harvest moonflowers from the Forbidden Forest. Without restrictions, the flowers will go extinct after this year. The table shows each prefect's annual benefits (in galleons), both under current harvesting and with proposed restrictions that would prevent extinction. The restrictions cost 60 galleons yearly, split equally (12 galleons each).

Benefits (in galleons)	Hermione	Draco	Hannah	Cho	Padma
Without restrictions	25	22	18	16	12
With restrictions (not including fee)	40	35	32	20	15
	<i>minus fee</i> 28	23	20	8	3
	✓	✓	✓	x	x

Q2.a | Hermione

For Hermione specifically:

- What are her benefits without restrictions? 25 *7 :: correct, 2 :: incorrect*
- What would her annual benefits be with restrictions (remember to subtract the fee)? 28 *7 :: correct, 2 :: incorrect*
- Based on this, would she vote for the restrictions? YES *7 :: correct, 6 :: incorrect, but follows from land 2, 2 :: incorrect*

Q2.b | Majority Vote

How many prefects will vote for the proposal? 3 *6 :: correct, 1 :: incorrect*

How many prefects will vote against the proposal? 2 *5 :: correct, 1 :: incorrect*

Q2.c | Benefits In Both Scenarios

What is the total annual benefit across all prefects:

- Without restrictions? 93 *93 = 25 + 22 + 18 + 16 + 12* *5 :: correct, 4 :: incorrect, but only a small math error, 1 :: incorrect process*
- With restrictions (remember to subtract the fee)? 82 *82 = 28 + 23 + 20 + 8 + 3* *5 :: correct, 4 :: incorrect, but only a small math error, 1 :: incorrect process*

Q2.d | Social Efficiency

Is implementing the harvesting restrictions with a 12 galleon fee socially efficient (i.e. have higher benefits)?

- No
- 10 :: correct, no, since benefits of no restrictions are greater than benefits with restrictions*
- 8 :: incorrect, but follows*
- 2 :: incorrect*